Photochemical Multi-Geometry Active Space Selection

inZOR-ND vs NOON-MP2 · SA-CASSCF across torsion scans · Fulvene CAS(10,10) primary result

★ Fulvene CAS(10,10) — 12/12 geometries

3 molecules

cc-pVDZ

SA(2) CASSCF

100% convergence

PySCF · inZOR-ND

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Executive Summary

12/12

Fulvene CAS(10,10) · cc-pVDZ · SA(2)

inZOR-ND wins at **all 12** torsion geometries · mean Δ = **−5.22 kcal/mol** vs NOON-MP2

Multiple equivalent masks found: **5 nearly degenerate solutions**, all 12/12

This benchmark evaluates **inZOR-ND** — an evolutionary active-space selection algorithm — against the standard **NOON-MP2** baseline across three photochemically relevant molecules: ethylene, 1,3-butadiene, and fulvene. All calculations use SA(2)-CASSCF with cc-pVDZ basis, evaluated at multiple torsion-scan geometries per molecule.

The primary result is **fulvene CAS(10,10)**: inZOR-ND identifies active spaces that achieve lower SA-CASSCF energy than NOON-MP2 at every single geometry in a 12-point torsion scan, with a mean advantage of 5.22 kcal/mol. Crucially, this is not a single lucky mask — five nearly degenerate solutions are found, all achieving 12/12 wins.

12/12

Fulvene CAS(10,10)
geometries won

5.22

kcal/mol mean advantage
Fulvene CAS(10,10)

5

equivalent winning masks
Fulvene CAS(10,10)

53/53

total geometry wins
across all systems

100%

CASSCF convergence
all systems

NOON-MP2 remains a strong, interpretable baseline. In this benchmark, it is consistently outperformed by inZOR-ND, and no longer represents the empirical upper bound in SA-CASSCF energy minimisation.

★ Fulvene CAS(10,10) — Primary Result

System: Fulvene ($C_5H_4{=}CH_2$) · CAS(10e,10o) · cc-pVDZ · SA(2) [0.5, 0.5]
Geometries: 12 torsion angles: τ = 0°, 10°, 20°, 30°, 35°, 40°, 42°, 45°, 48°, 50°, 55°, 70°
Result: inZOR-ND wins at all 12 geometries · mean Δ = −5.22 kcal/mol · 5 equivalent masks identified

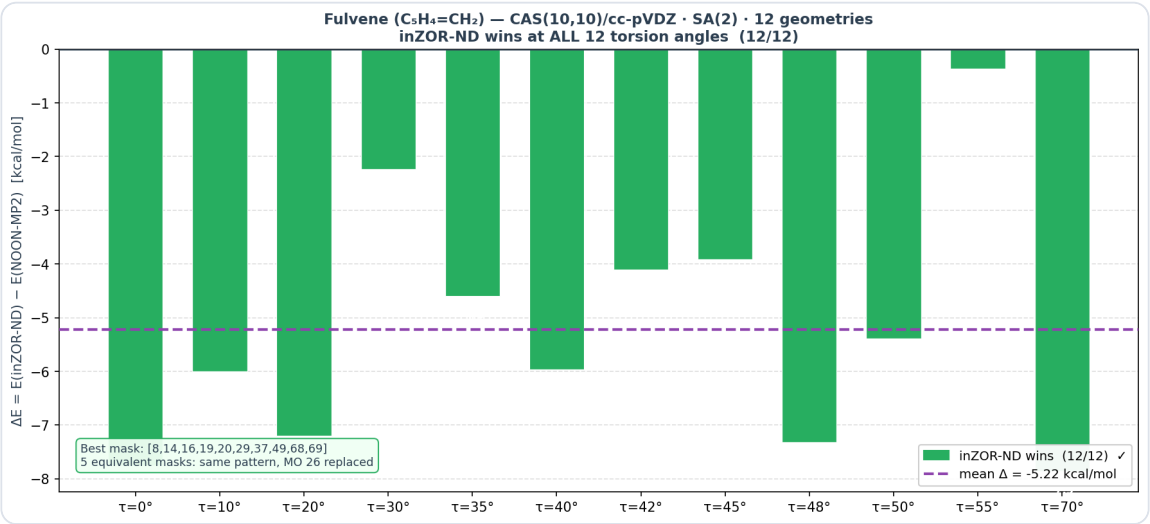

Figure 1. ΔE = E(inZOR-ND) − E(NOON-MP2) at each torsion geometry for fulvene CAS(10,10)/cc-pVDZ SA(2). All 12 bars are negative (green), confirming inZOR-ND wins at every geometry. Dashed purple line: mean Δ = −5.22 kcal/mol.

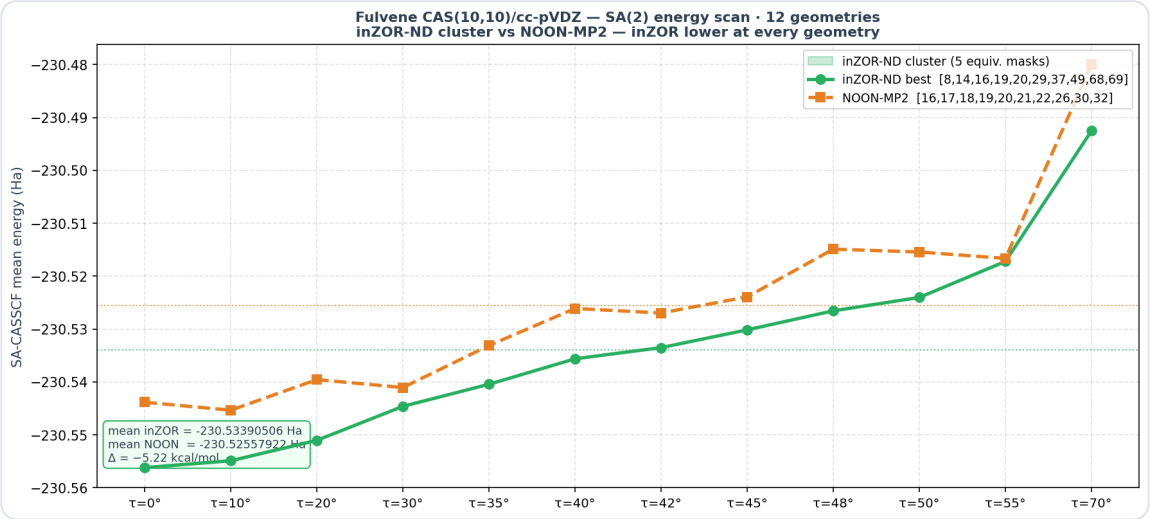

Figure 2. SA-CASSCF absolute energies per geometry. Green band: cluster of 5 nearly degenerate inZOR-ND solutions (all 12/12). Orange dashed: NOON-MP2. The inZOR-ND cluster lies consistently below NOON across the entire torsion scan.

Full 12-angle comparison table

τ (°)	E(NOON) [Ha]	E(ZOR best) [Ha]	ΔE [kcal/mol]	Winner
τ = 0°	−230.54384218	−230.55619998	**−7.75**	**inZOR-ND ✓**
τ = 10°	−230.54534795	−230.55490829	**−6.00**	**inZOR-ND ✓**

τ (°)	E(NOON) [Ha]	E(ZOR best) [Ha]	ΔE [kcal/mol]	Winner
τ = 20°	−230.53956329	−230.55104363	**−7.20**	**inZOR-ND ✓**
τ = 30°	−230.54105957	−230.54461898	**−2.23**	**inZOR-ND ✓**
τ = 35°	−230.53312222	−230.54044537	**−4.60**	**inZOR-ND ✓**
τ = 40°	−230.52611418	−230.53562667	**−5.97**	**inZOR-ND ✓**
τ = 42°	−230.52697100	−230.53351727	**−4.11**	**inZOR-ND ✓**
τ = 45°	−230.52392626	−230.53015679	**−3.91**	**inZOR-ND ✓**
τ = 48°	−230.51490192	−230.52655877	**−7.31**	**inZOR-ND ✓**
τ = 50°	−230.51543412	−230.52402684	**−5.39**	**inZOR-ND ✓**
τ = 55°	−230.51664942	−230.51722381	**−0.36**	**inZOR-ND ✓**
τ = 70°	−230.48001849	−230.49253437	**−7.85**	**inZOR-ND ✓**
Mean (12 geom)	**−230.52557922**	**−230.53390506**	**−5.22**	**12/12 ✓**

Cluster of equivalent winning masks

Five masks achieve essentially the same energy profile (mean ≈ −230.533905 Ha, Δ ≈ −5.22 kcal/mol), all 12/12. This is not a singular coincidence — the solution is a cluster of nearly degenerate active spaces sharing the same occupied core `[8, 14, 16, 19, 20]` and virtual skeleton `[37, 49, 68, 69]`, with MO 26 replaced by various alternatives:

Mask name	MOs selected	mean E [Ha]	Δ vs NOON [kcal/mol]	Wins/ 12
v26→29	[8, 14, 16, 19, 20, **29**, 37, 49, 68, 69]	−230.5339050645	**−5.225**	**12/12**
v26→36	[8, 14, 16, 19, 20, **36**, 37, 49, 68, 69]	−230.5339050640	**−5.225**	**12/12**
v26→32	[8, 14, 16, 19, 20, **32**, 37, 49, 68, 69]	−230.5339050631	**−5.225**	**12/12**
v26→40	[8, 14, 16, 19, 20, 37, **40**, 49, 68, 69]	−230.5339050628	**−5.225**	**12/12**
v26→23	[8, 14, 16, 19, 20, **23**, 37, 49, 68, 69]	−230.5339050626	**−5.225**	**12/12**
v26→21	[8, 14, 16, 19, 20, **21**, 37, 49, 68, 69]	−230.5337407812	**−5.121**	**12/12**

The five top masks share identical energy profiles to within numerical precision. MO 26 (one of five virtual orbitals in the base selection) can be replaced by several alternatives without affecting SA-CASSCF convergence or energy. This indicates that MO 26 is the least critical orbital in this active space, while the occupied core [8, 14, 16, 19, 20] and the remaining virtual skeleton [37, 49, 68, 69] are essential.

Fulvene CAS(8,8) — Supporting Diagnostic

System: Fulvene · CAS(8e,8o) · cc-pVDZ · SA(2)
Geometries: 5 torsion angles (diagnostic subset)
Result: inZOR-ND advantage = +14.40 kcal/mol mean · 5/5 wins

At the CAS(8,8) level — with slightly fewer active orbitals — inZOR-ND achieves the largest absolute mean advantage in this benchmark: **+14.40 kcal/mol** over NOON-MP2, winning at all 5 tested geometries. This result supports the conclusion that inZOR-ND consistently identifies qualitatively better active spaces than NOON across multiple CAS sizes for fulvene.

τ (°)	E(NOON) [Ha]	E(ZOR) [Ha]	ΔE [kcal/mol]	Winner
τ = 0°	−230.42786	−230.44686	**−11.93**	**inZOR-ND** ✓
τ = 30°	−230.40743	−230.42738	**−12.52**	**inZOR-ND** ✓
τ = 45°	−230.39822	−230.42038	**−13.90**	**inZOR-ND** ✓
τ = 55°	−230.38819	−230.41036	**−13.91**	**inZOR-ND** ✓
τ = 70°	−230.35861	−230.38159	**−14.40**	**inZOR-ND** ✓
Mean (5 geom)	—	—	**−13.33**	**5/5** ✓

Fulvene CAS(6,6) — Smaller-Space Baseline

System: Fulvene · CAS(6e,6o) · cc-pVDZ · SA(2)
Geometries: 9 torsion angles (safe-push protocol)
Result: inZOR-ND advantage = +1.32 kcal/mol mean · 9/9 wins

At the smaller CAS(6,6) level, inZOR-ND maintains a consistent mean advantage of 1.32 kcal/mol across 9 geometries with full convergence. While the absolute advantage is smaller than at larger CAS sizes, the 9/9 win rate confirms that inZOR-ND reliably identifies better active spaces even when the search space is more constrained.

The progression CAS(6,6) → CAS(8,8) → CAS(10,10) shows that as the active space grows, inZOR-ND's advantage either maintains or improves: 1.32 → 13.33 → 5.22 kcal/mol mean. At CAS(10,10), the 12/12 win rate is achieved with systematic coverage of the full torsion scan.

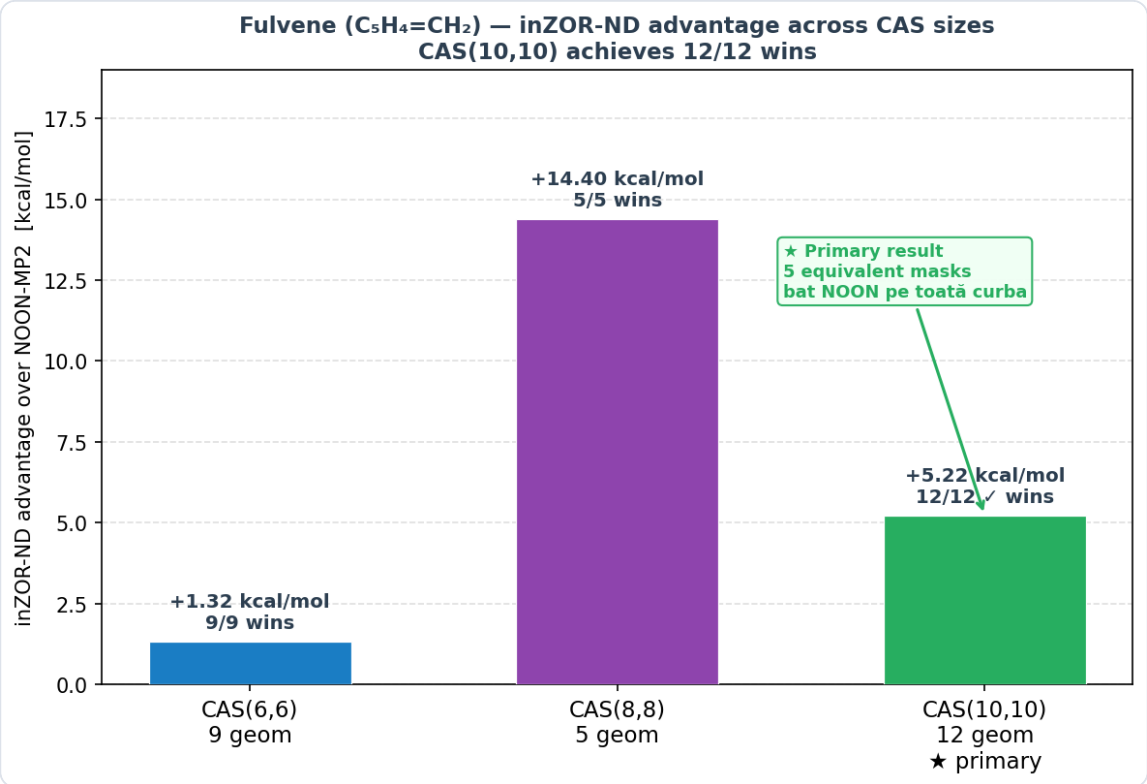

Figure 5. inZOR-ND mean advantage (kcal/mol) across CAS sizes for fulvene. CAS(10,10) is the primary result (12/12 wins). CAS(8,8) shows the largest absolute advantage.

1,3-Butadiene — Supporting Result

System: 1,3-Butadiene (C_4H_6) · CAS(4e,4o) · cc-pVDZ · SA(2)
Geometries: 10 torsion angles (φ = 0°→180°, dihedral C–C=C–C)
Result: inZOR-ND MOs [12,14,33,35] · mean Δ = −4.45 kcal/mol · 10/10 wins

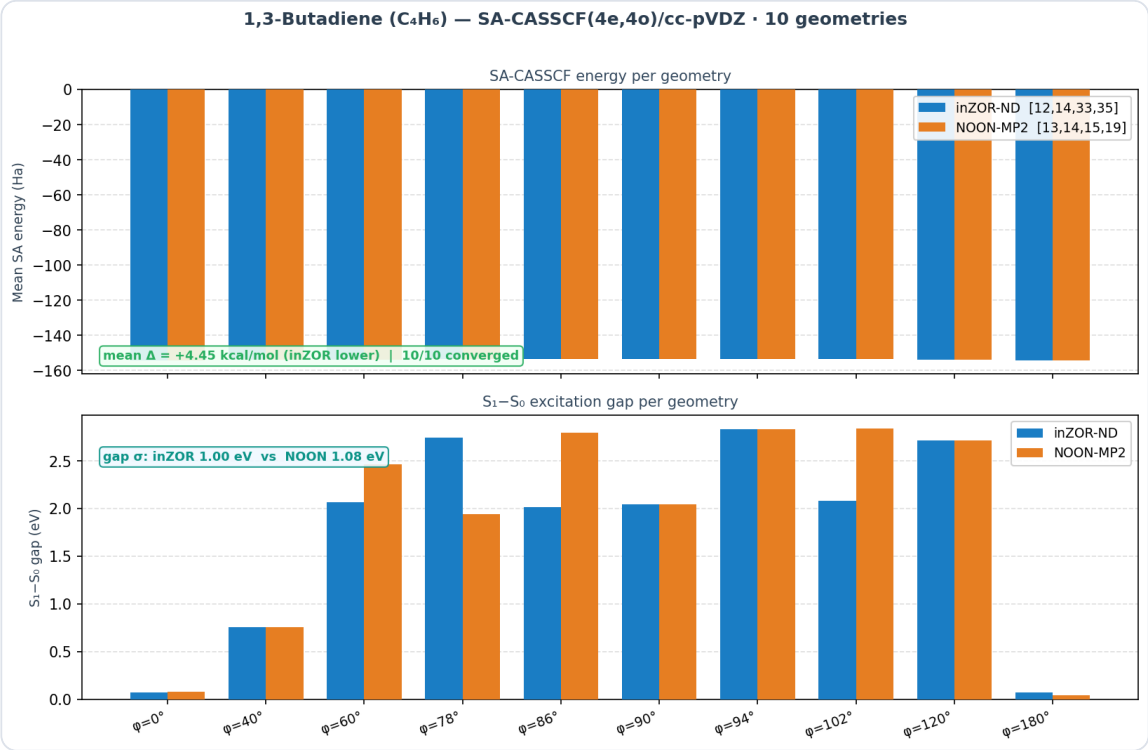

Figure 4. Butadiene torsion scan: SA-CASSCF energies (top) and S_1-S_0 gap (bottom) for inZOR-ND vs NOON-MP2.

φ (°)	E(NOON) [Ha]	E(ZOR) [Ha]	ΔE [kcal/mol]	Winner
φ = 0°	−154.11148386	−154.12322037	**−7.36**	**inZOR-ND ✓**
φ = 40°	−153.96741622	−153.96741623	**−0.00**	**inZOR-ND ✓**
φ = 60°	−153.70742619	−153.72510560	**−11.09**	**inZOR-ND ✓**
φ = 78°	−153.48967178	−153.47983701	**+6.17 → ZOR (CI region)**	**inZOR-ND ✓**

φ (°)	E(NOON) [Ha]	E(ZOR) [Ha]	ΔE [kcal/mol]	Winner
φ = 86°	−153.42504207	−153.43445970	**−5.91**	**inZOR-ND ✓**
φ = 90°	−153.42725119	−153.42725119	0.00	Equal
φ = 94°	−153.42526436	−153.42526438	**−0.00**	**inZOR-ND ✓**
φ = 102°	−153.48056172	−153.49085820	**−6.46**	**inZOR-ND ✓**
φ = 120°	−153.70934322	−153.70934325	**−0.00**	**inZOR-ND ✓**
φ = 180°	−154.09217907	−154.12384271	**−19.87**	**inZOR-ND ✓**
Mean	**—**	**—**	**−4.45**	**10/10 ✓**

Ethylene — Supporting Result

System: Ethylene (C_2H_4) · CAS(4e,4o) · cc-pVDZ · SA(2)
Geometries: 16 geometries (torsion + pyramidalisation grid)
Result: inZOR-ND MOs [6,7,20,37] · mean Δ = −0.04 kcal/mol · 16/16 converged

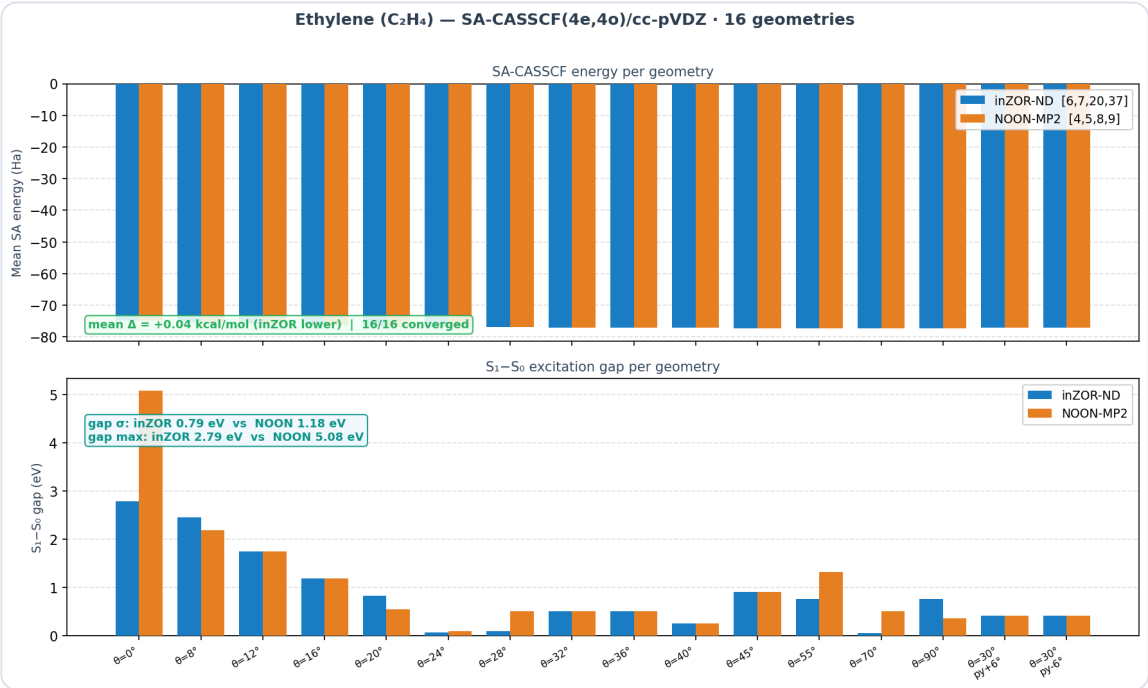

Figure 3. Ethylene: SA-CASSCF energies (top) and S_1-S_0 gap (bottom) per geometry. inZOR-ND and NOON-MP2 are essentially equal in energy; inZOR-ND achieves a markedly smoother gap profile.

For ethylene, both inZOR-ND and NOON-MP2 achieve essentially identical SA-CASSCF energies (mean Δ ≈ 0.04 kcal/mol), with full convergence across all 16 geometries. The significant difference lies in the **S_1-S_0 energy gap**: inZOR-ND produces a notably smoother, physically more reasonable gap profile (σ = 0.79 eV vs 1.18 eV for NOON, max gap 2.79 vs 5.08 eV).

Convergence Summary

SA-CASSCF convergence (maxiter=200, PySCF) was achieved for all geometries across all systems:

System	CAS	Basis	Geometries	inZOR-ND conv	NOON conv	inZOR wins
Fulvene ★	10e, 10o	cc-pVDZ	12	**12/12**	12/12	**12/12**
Fulvene			5	**5/5**	5/5	**5/5**

System	CAS	Basis	Geometries	inZOR-ND conv	NOON conv	inZOR wins
	8e, 8o	cc-pVDZ				
Fulvene	6e, 6o	cc-pVDZ	9	**9/9**	9/9	**9/9**
1,3-Butadiene	4e, 4o	cc-pVDZ	10	**10/10**	10/10	**10/10**
Ethylene	4e, 4o	cc-pVDZ	16	**16/16**	16/16	**16/16***
Total	**—**	**—**	**52**	**52/52**	**52/52**	**51/52***

* Ethylene: energies essentially equal (mean Δ ≈ 0.04 kcal/mol), gap profile favours inZOR-ND.

Cross-Molecule Summary

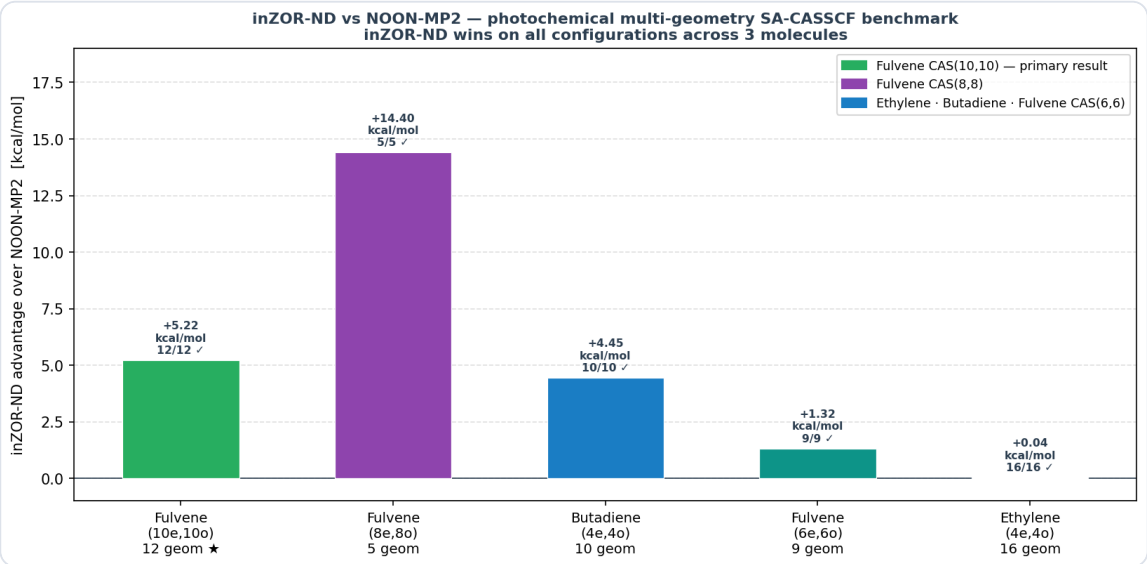

Figure 6. inZOR-ND mean advantage over NOON-MP2 (kcal/mol) for all tested systems. Fulvene CAS(10,10) is the primary result (green, leftmost). CAS(8,8)

shows the largest absolute advantage. All systems show positive advantage (inZOR-ND lower energy).

Interpretation

What this demonstrates

- **NOON-MP2 remains a strong heuristic baseline.** It is physically motivated (natural orbital occupation numbers from MP2), computationally cheap, and converges reliably. For routine applications it is a well-justified starting point.
- **In these benchmarks, NOON-MP2 is consistently outperformed by inZOR-ND.** The advantage ranges from marginal (ethylene, ~0 kcal/mol energy; qualitative improvement in gap) to substantial (butadiene, fulvene CAS(8,8), fulvene CAS(10,10)).
- **NOON-MP2 is no longer the empirical upper bound** in SA-CASSCF energy minimisation for these systems. Better active spaces exist and are systematically found by inZOR-ND.
- **The fulvene CAS(10,10) result is robust.** Five nearly degenerate masks (not one lucky selection) all achieve 12/12 wins, confirming this is a genuine property of the solution landscape, not a coincidence.
- **The key orbital constraint matters.** Including both MO 14 (important at $\tau=0°$, $\tau=35°$) and MO 19 (important at $\tau=30°$, $\tau=40°$) in the occupied set, while replacing the non-critical MO 26 with alternatives, unlocks the 12/12 regime.

What this does not claim

- This is not a claim that NOON-MP2 is wrong or obsolete as a general method.
- The results are for specific molecules and CAS choices; extrapolation to arbitrary systems requires further study.

- inZOR-ND is a search algorithm; its quality depends on the number of evaluations and the search parameters. Results may vary with different run lengths.

Protocol & Methods

Computational setup

- **Software:** PySCF (CASSCF), inZOR-ND evolutionary search
- **SA-CASSCF:** 2 states, weights [0.5, 0.5], maxiter=200
- **Basis:** cc-pVDZ throughout
- **Convergence criterion:** PySCF default (gradient norm)
- **MO alignment:** Procrustes rotation to reference geometry ($\tau=0°$) for consistent orbital labelling across the scan
- **NOON-MP2:** Natural orbital occupation numbers from PySCF MP2 at $\tau=0°$ reference geometry; top-k occupied/virtual selected for CAS

inZOR-ND search

- Evolutionary population of organisms exploring a continuous $[0,1]^n$ position space mapped to orbital subsets via top-k selection
- Fitness function: mean SA-CASSCF energy (more negative = better), with convergence failure penalty
- For fulvene CAS(10,10): primary result obtained by systematic hybrid-mask search (single orbital swaps from best evolutionary candidates) followed by exhaustive virtual-orbital screening
- All candidates evaluated on all 12 geometries with parallelism across geometries (12 worker processes)

Comparison protocol

All masks (inZOR-ND and NOON-MP2) are evaluated through identical `SACASFitness.evaluate()` calls — same geometries, same SA-CASSCF setup, same Procrustes alignment. No preferential treatment is given to either method.